M201103W- Signals and Systems

ssch7_4 –Properties of unilateral z-transform

14:10 pm Wednesday @ C107, 1st Teaching Building

14:10 pm Friday @ A107, 1st Teaching Building

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Properties of unilateral *z*-transform

- Linearity
- Shifting
- Convolution
- Summation
- Exponential weighting

  (Scaling in the z-domain)
- Differentiation in the z-domain
- Time Reversal
- Initial- and final- value theorems

Properties of unilateral *z*-transform

► Exponential weighting(Scaling in the z-domain)

If $$x[k]u[k] \xleftrightarrow{\mathscr{Z}} X(z), \qquad |z| > R_x$$

Then $$a^k x[k]u[k] \xleftrightarrow{\mathscr{Z}} X(z/a), \qquad |z| > |a|R_x$$

$$\mathscr{Z}\{a^k x[k]u[k]\} = \sum_{k=0}^{+\infty} a^k x[k] z^{-k} = \sum_{k=0}^{+\infty} x[k](z/a)^{-k} = X(z/a),$$

If $x[k]$ is weighted with a^k, then z-transform will scale a times.

Example. Calculate the z transform of $\alpha^k \cos(\Omega_0 k)u[k]$

Solution:

Because $\cos(\Omega_0 k)u[k] \xleftrightarrow{\mathcal{Z}} \dfrac{1 - z^{-1}\cos\Omega_0}{1 - 2z^{-1}\cos\Omega_0 + z^{-2}}, \quad |z| > 1$

According to exponential weighting property:

$$\alpha^k \cos(\Omega_0 k)u[k] \xleftrightarrow{\mathcal{Z}} \frac{1 - (z/\alpha)^{-1}\cos\Omega_0}{1 - 2(z/\alpha)^{-1}\cos\Omega_0 + (z/\alpha)^{-2}}$$

$$= \frac{1 - \alpha z^{-1}\cos\Omega_0}{1 - 2\alpha z^{-1}\cos\Omega_0 + \alpha^2 z^{-2}}, \quad |z| > |\alpha|$$

Properties of unilateral *z*-transform

► Differentiation in the z-domain

If: $$x[k]u[k] \xleftrightarrow{\mathcal{Z}} X(z), \qquad |z| > R_x$$

thus $$kx[k]\,u[k] \xleftrightarrow{\mathcal{Z}} -z\frac{\mathrm{d}X(z)}{\mathrm{d}z}, \quad |z| > R_x$$

$$X(z) = \sum_{k=0}^{+\infty} x[k]\,u[k]\,z^{-k} \qquad \frac{\mathrm{d}X(z)}{\mathrm{d}z} = -z^{-1}\cdot\sum_{k=0}^{+\infty} kx[k]\,u[k]z^{-k},$$

If $x[k]$ is multiplied by k, the new z-transform is the differentiation of $X(z)$ multiplied by $(-z)$.

Example. Calculate the unilateral z-transform of $x[k]=ka^k u[k]$. Determine the ROC.

Solution: $$a^k u[k] \xleftrightarrow{\mathscr{Z}} \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \qquad |z|>|a|$$

According to differentiation property in z-Transform,

$$ka^k u[k] \xleftrightarrow{\mathscr{Z}} -z\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{z}{z-a}\right)$$

$$= \frac{az}{(z-a)^2} = \frac{az^{-1}}{(1-az^{-1})^2}, \quad |z|>|a|$$

Properties of unilateral *z*-transform

► Time reversal property

If: $$x[k]u[k] \xleftrightarrow{\mathcal{Z}} X(z), \qquad |z| > R_x$$

thus $$x[-k] \xleftrightarrow{\mathcal{Z}} X(1/z) \qquad \frac{1}{R_{x+}} < |z| < \frac{1}{R_{x-}}$$

Properties of unilateral *z*-transform

- Initial- and final- value theorems (X(z) is given, calculate $x[0]$ and $x[+\infty]$)

If $$x[k]u[k] \xleftrightarrow{\mathscr{Z}} X(z) \qquad |z| > R_x$$

then $$x[0] = \lim_{z \to +\infty} X(z)$$

If ROC of $(z-1)X(z)$ includes the unit circle,

$$x[+\infty] = \lim_{z \to 1} (z-1) X(z)$$

Properties of unilateral *z*-transform

- Initial- and final- value theorems

$$x[0] = \lim_{z \to +\infty} X(z)$$

Proof: $X(z) = \sum_{k=0}^{+\infty} x[k]z^{-k} = x[0] + x[1]z^{-1} + x[2]z^{-2} + x[3]z^{-3} + \cdots$

When $z \to +\infty$, $x[0]$ is left on the right side of the equation above.

$$x[0] = \lim_{z \to +\infty} X(z)$$

Properties of unilateral *z*-transform

- Initial- and final- value theorems

$$x[+\infty] = \lim_{z \to 1} (z-1) X(z)$$

Proof: $$\sum_{k=0}^{+\infty} \left(x[k+1] - x[k] \right) z^{-k} = \{ zX(z) - x[0] \} - X(z) = (z-1)X(z) - x[0]$$

If z=1 :

$$\sum_{k=0}^{+\infty} \left(x[k+1] - x[k] \right) = (x[1] - x[0]) + (x[2] - x[1]) + (x[3] - x[2]) + \ldots + (x[+\infty] - x[+\infty - 1])$$

$$= x[+\infty] - x[0]$$

Therefore: $$x[+\infty] = \lim_{z \to 1} (z-1) X(z)$$

Example.

If $X(z) = 1/(1-a\,z^{-1})$, $|z| > |a|$, calculate $x[0]$, $x[1]$ and $x[+\infty]$.

$$x[0] = \lim_{z \to +\infty} X(z)$$

$$x[+\infty] = \lim_{z \to 1} (z-1) X(z)$$

Verification: $X(z) = 1/(1-a\,z^{-1})$, $|z| > |a|$

In the time domain, $x[k] = a^k u[k]$

$x[0]=1$, $x[1]= a$, $x[+\infty]=0$